

CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

## Chapter 5: Number Play

Topic-wise short notes with solved examples

Parity, multiples and remainders studied with algebra; why the divisibility tests for 3, 9 and 11 work; digital roots; and cryptarithms.

### 1 Parity

- Even numbers have the form  $2n$  and odd numbers the form  $2n + 1$ . Negative numbers such as  $-4$  are also even.
- $\text{even} \pm \text{even} = \text{even}$ ;  $\text{odd} \pm \text{odd} = \text{even}$ ;  $\text{even} \pm \text{odd} = \text{odd}$ .
- $a + b$  and  $a - b$  always have the **same parity**, because they differ by  $2b$ .
- So for any numbers  $a, b, c, d$ , all the expressions  $a \pm b \pm c \pm d$  have the same parity. Changing one sign changes the value by an even number.
- For four **consecutive** numbers,  $a \pm b \pm c \pm d$  is always even, because two of the numbers are odd and two are even.
- An expression is always even if 2 can be taken out as a factor:  $4m + 2q = 2(2m + q)$ .

**Example 1.1** Which of these are even for all integer values of the letters:  $6m - 4n$ ,  $3g + 5h$ ,  $x^2 + 2$ ,  $13k - 5k$ ?

SOLUTION

$6m - 4n = 2(3m - 2n)$ : always even.

$3g + 5h$ : odd when  $g = 1, h = 0$ , so not always even.

$x^2 + 2$ : odd when  $x$  is odd.

$13k - 5k = 8k$ : always even.

**$6m - 4n$  and  $13k - 5k$**

**Example 1.2** Place  $+$  or  $-$  signs between 5, 6, 7, 8. Can the result be 3?

SOLUTION

$5 + 6 + 7 + 8 = 26$  is even, and every other choice of signs has the same parity.

**No. Every result is even.**

### 2 Sums of consecutive numbers

- Every odd number is a sum of two consecutive numbers:  $2n + 1 = n + (n + 1)$ .
- The sum of three consecutive numbers is 3 times the middle one:  $(n - 1) + n + (n + 1) = 3n$ .
- The sum of four consecutive numbers is  $4n + 6$ , which is even but never a multiple of 4.

**Example 2.1** The sum of four consecutive numbers is 58. Find them.

SOLUTION

$$n + (n + 1) + (n + 2) + (n + 3) = 4n + 6 = 58, \text{ so } n = 13$$

**13, 14, 15, 16**

**Example 2.2** The greatest of five consecutive numbers is  $p$ . Write their sum.

SOLUTION

The numbers are  $p - 4, p - 3, p - 2, p - 1, p$ .

**$5p - 10$**

### 3 Even numbers and multiples of 4

- An even number is either a multiple of 4 (form  $4p$ ) or 2 more than a multiple of 4 (form  $4p + 2$ ).

| Numbers added         | Sum            | Multiple of 4? |
|-----------------------|----------------|----------------|
| $4p$ and $4q$         | $4(p + q)$     | Yes            |
| $4p + 2$ and $4q + 2$ | $4(p + q + 1)$ | Yes            |
| $4p$ and $4q + 2$     | $4(p + q) + 2$ | No             |

- This is just like parity: two of the same kind give a multiple of 4, two of different kinds do not.

### 4 Properties of divisibility

- If  $a$  divides  $M$  and  $a$  divides  $N$ , then  $a$  divides  $M + N$  and  $M - N$ .
- If  $A$  is divisible by  $k$ , then every multiple of  $A$  is divisible by  $k$ .
- If  $A$  is divisible by  $k$ , then  $A$  is divisible by every factor of  $k$ .
- If  $A$  is divisible by  $k$  and by  $m$ , then  $A$  is divisible by the LCM of  $k$  and  $m$  (not always by  $k \times m$ ).
- The reverse statements can fail: a multiple of 8 can be a sum of two numbers that are not multiples of 8 ( $72 = 50 + 22$ ); a multiple of 7 need not be a multiple of 28.

**Example 4.1** Always, sometimes or never true: "If a number is divisible by both 6 and 4, it is divisible by 24."

SOLUTION

The LCM of 6 and 4 is 12. So the number is certainly divisible by 12.

12 is divisible by 6 and 4 but not by 24. 48 is divisible by all three.

**Sometimes true**

**Example 4.2** Always, sometimes or never true: "The sum of a multiple of 6 and a multiple of 9 is a multiple of 3."

SOLUTION

$$6x + 9y = 3(2x + 3y)$$

**Always true**

**Example 4.3** Always, sometimes or never true: "An odd number plus an even number is a multiple of 6."

SOLUTION

The sum is odd, but every multiple of 6 is even.

**Never true**

## 5 Remainders

- Numbers that leave remainder  $r$  on division by  $n$  have the form  $nk + r$ . Those leaving remainder 3 on division by 5 are  $5k + 3$ : 3, 8, 13, 18, ... The same numbers can be written  $5k - 2$  with  $k \geq 1$ .
- Remainders can be added, subtracted and multiplied.** If two numbers leave remainders  $r$  and  $s$  on division by  $n$ , their sum, difference and product leave the same remainders as  $r + s$ ,  $r - s$  and  $rs$  (reduced by multiples of  $n$ ).
- If a number leaves the same remainder  $r$  with several divisors, it has the form (LCM of the divisors)  $\times k + r$ .

**Example 5.1** On division by 7, the number 661 leaves remainder 3 and 4779 leaves remainder 5. Find the remainders of  $4779 + 661$  and  $4779 - 661$ .

SOLUTION

$$\text{Write } 4779 = 7p + 5 \text{ and } 661 = 7q + 3.$$

$$\text{Sum} = 7(p + q) + 8 = 7(p + q + 1) + 1. \text{ Difference} = 7(p - q) + 2.$$

**Remainders 1 and 2**

**Example 5.2** Find the smallest number that leaves remainder 2 on division by 3, remainder 3 on division by 4 and remainder 4 on division by 5.

SOLUTION

Each remainder is 1 less than the divisor, so the number is 1 less than a common multiple of 3, 4 and 5.

$$\text{LCM}(3, 4, 5) = 60$$

**59**

**Example 5.3** Describe all numbers that leave remainder 2 on division by 3 and also on division by 4.

SOLUTION

Such a number is 2 more than a common multiple of 3 and 4.

$12n + 2$ : 2, 14, 26, 38, ...

## 6 Divisibility by 2, 4, 5, 8 and 10

- A number ...  $dcb a$  equals ...  $+ 1000d + 100c + 10b + a$ . Every place value except the units is a multiple of 10 (and so of 2 and 5).

| Divisor | Test                                                                                     |
|---------|------------------------------------------------------------------------------------------|
| 10      | Units digit is 0                                                                         |
| 5       | Units digit is 0 or 5                                                                    |
| 2       | Units digit is even                                                                      |
| 4       | Number formed by the last two digits is divisible by 4 (since 100 is a multiple of 4)    |
| 8       | Number formed by the last three digits is divisible by 8 (since 1000 is a multiple of 8) |

## 7 Divisibility by 9 and 3

- Each power of 10 is 1 more than a multiple of 9:  $10 = 9 + 1$ ,  $100 = 99 + 1$ ,  $1000 = 999 + 1$ .
- So a number and the **sum of its digits** leave the same remainder on division by 9 (and by 3).
- A number is divisible by 9 exactly when its digit sum is divisible by 9, and by 3 exactly when its digit sum is divisible by 3.
- Rearranging the digits does not change the digit sum, so it does not change divisibility by 9 or 3.

**Example 7.1** Find the remainder when 7309 is divided by 9.

SOLUTION

$7 + 3 + 0 + 9 = 19$ , then  $1 + 9 = 10$ , then  $1 + 0 = 1$

1

**Example 7.2**  $31z5$  is a multiple of 9. Find the digit  $z$ .

SOLUTION

Digit sum  $= 9 + z$  must be a multiple of 9.

$z = 0$  or  $z = 9$

**Example 7.3** How many multiples of 9 lie between 4300 and 4400?

SOLUTION

The first is 4302 (digit sum 9) and the last is 4392 (digit sum 18).

$$\frac{4392 - 4302}{9} + 1$$

11

## 8 Divisibility by 11

- Place values are alternately 1 more and 1 less than a multiple of 11:  $1 = 0 + 1$ ,  $10 = 11 - 1$ ,  $100 = 99 + 1$ ,  $1000 = 1001 - 1$ .
- Test:** starting from the units digit, put alternate + and – signs before the digits and add. The result has the same remainder as the number on division by 11. The number is divisible by 11 exactly when the result is 0 or a multiple of 11.
- If the result is negative, add 11 to get the remainder.

**Example 8.1** Is 90904 divisible by 11?

SOLUTION

From the units digit:  $4 - 0 + 9 - 0 + 9 = 22$

**Yes, since 22 is a multiple of 11**

**Example 8.2** Find the remainder when 5529 is divided by 11.

SOLUTION

$$9 - 2 + 5 - 5 = 7$$

7

**Example 8.3** Find the remainder when 328105 is divided by 11.

SOLUTION

$$5 - 0 + 1 - 8 + 2 - 3 = -3, \text{ and } -3 + 11 = 8$$

8

## 9 Divisibility by other numbers

- To test divisibility by a product, use factors that have **no common factor**: 6 uses 2 and 3; 12 uses 3 and 4; 15 uses 3 and 5; 18 uses 2 and 9; 24 uses 3 and 8; 36 uses 4 and 9; 44 uses 4 and 11.
- Testing 24 with 4 and 6 fails, because the LCM of 4 and 6 is only 12. The number 12 passes both tests but is not divisible by 24.

**Example 9.1**  $48a23b$  is a multiple of 18. Find all pairs of digits  $(a, b)$ .

SOLUTION

Divisible by 2:  $b$  is 0, 2, 4, 6 or 8.

Divisible by 9:  $17 + a + b$  is a multiple of 9, so  $a + b = 1$  or  $a + b = 10$ .

$(1, 0), (8, 2), (6, 4), (4, 6), (2, 8)$

**Example 9.2**  $3p7q8$  is divisible by 44. Find all pairs  $(p, q)$ .

SOLUTION

Divisible by 4:  $q8$  is a multiple of 4, so  $q$  is 0, 2, 4, 6 or 8.

Divisible by 11:  $(8 + 7 + 3) - (p + q) = 18 - (p + q)$  is 0 or 11. Since  $p + q = 18$  needs  $q = 9$ , we get  $p + q = 7$ .

$(7, 0), (5, 2), (3, 4), (1, 6)$

## 10 Digital roots

- Add the digits of a number, then the digits of the result, and so on until one digit is left. That digit is the **digital root**.
- The digital root is the remainder on division by 9, except that multiples of 9 have digital root 9.
- Digital roots of consecutive numbers run  $1, 2, \dots, 9, 1, 2, \dots$ . Adding 9 to a number does not change its digital root.
- Digital roots can check calculations: the digital root of a sum (or product) equals the digital root of the sum (or product) of the digital roots. This check was described by Āryabhaṭa II (about 950 CE).

**Example 10.1** Find the digital root of 489710.

SOLUTION

$4 + 8 + 9 + 7 + 1 + 0 = 29, 2 + 9 = 11, 1 + 1 = 2$

**2**

**Example 10.2** What is the digital root of  $9a + 36b + 13$ ?

SOLUTION

$9a + 36b + 13 = 9(a + 4b + 1) + 4$

**4**

**Example 10.3** Check  $234 \times 56 = 13104$  using digital roots.

SOLUTION

Digital roots: 234 gives 9, 56 gives 2. Product of roots = 18, digital root 9.

13104 has digit sum 9.

**The roots agree, so the answer passes the check.**

## 11 Cryptarithms

- In a cryptarithm each letter stands for one digit, different letters stand for different digits, and a number never starts with 0.
- Useful clues: the units digits, carries, the number of digits in the answer, and parity.

**Example 11.1** Solve  $AB + 37 = 6A$ .

SOLUTION

Units:  $B + 7$  ends in  $A$ . Tens:  $A + 3$  plus any carry is 6, so  $A = 3$  with no carry or  $A = 2$  with a carry.

If  $A = 3$ ,  $B + 7$  ends in 3 gives  $B = 6$ , which produces a carry: not allowed. If  $A = 2$ ,  $B + 7 = 12$  gives  $B = 5$ .

$$25 + 37 = 62$$

**Example 11.2** Solve  $PQ \times 8 = RS$ .

SOLUTION

The product has two digits, so  $PQ$  is at most 12.  $PQ = 10$  gives 80 (repeats 0),  $PQ = 11$  repeats a digit.

$$12 \times 8 = 96$$

**Example 11.3** Solve  $JK \times 6 = KKK$ .

SOLUTION

$KKK = 111 \times K = 37 \times 3 \times K$ , so  $JK \times 2 = 37 \times K$ .  $K$  must be even.

$K = 2$  gives  $JK = 37$  (units digit not 2).  $K = 4$  gives  $JK = 74$ .

$$74 \times 6 = 444$$

## 12 Common mistakes

- Saying "divisible by 4 and 6 means divisible by 24". Use the LCM, which is 12.
- Adding digits to test divisibility by 11. That test uses **alternating** signs.
- Using the units digit to test divisibility by 3 or 9.
- Giving digital root 0 for a multiple of 9. It is 9.
- Deciding a statement is always true from a few examples. Use algebra, or find one counter-example.

MathSetu